UMD SCHOOL OF PUBLIC HEALTH TEACHING DEMONSTRATION

The Binomial Distribution

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Slides available at **thevazquezlab.com/talks**

Today's goals

By the end of class, you should be able to:

1. Recognize when a count follows a binomial distribution.
2. Compute binomial probabilities and the expected count np.
3. Use binomial reasoning to estimate the size of a population that can be hard to count.

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Running example

A clinic tests $n = 4$ patients for flu. Each patient tests positive (success) with chance $p = 0.25$, independently of the others. X = number of positive tests.
(Illustrative numbers.)

Building the formula, one piece at a time

Clinic: $n = 4$ patients, $p = 0.25$. What is the chance of **exactly one** positive test, $P(X = 1)$?

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1. **One arrangement.** Patient 1 positive, the other three negative: $+ \; - \; - \; -$

$$0.25 \times 0.75 \times 0.75 \times 0.75 = \underbrace{(0.25)^1}_{p^k} \; \underbrace{(0.75)^3}_{(1-p)^{n-k}} = 0.105$$

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2. **How many arrangements?** $+\ -\ -\ -$ $\quad -\ +\ -\ -$ $\quad -\ -\ +\ -$ $\quad -\ -\ -\ +$

$$\binom{4}{1} = \frac{4!}{1!\,3!} = \frac{4 \times 3 \times 2 \times 1}{1 \times (3 \times 2 \times 1)} = 4$$

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3. **Add them up.** Each arrangement has the same chance, so

$$P(X = 1) = 4 \times 0.105 = 0.422$$

The binomial formula

$$P(X = k) = \underbrace{\binom{n}{k}}_{\text{arrangements}} \underbrace{p^k}_{k \text{ successes}} \underbrace{(1-p)^{n-k}}_{n-k \text{ failures}}, \qquad k = 0, 1, \ldots, n$$

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}, \qquad n! = n \times (n-1) \times \cdots \times 1, \qquad 0! = 1$$

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Mean and variance

$$E(X) = np, \qquad \mathrm{Var}(X) = np(1-p)$$

Clinic: $E(X) = 4 \times 0.25 = 1, \qquad \mathrm{Var}(X) = 4 \times 0.25 \times 0.75 = 0.75$

Adding up: the cumulative distribution

The **cumulative distribution function** (CDF) adds the probabilities up to x:

$$F(x) = P(X \leq x) = \sum_{k=0}^{x} \binom{n}{k} p^k (1-p)^{n-k}$$

k	$P(X = k)$	$P(X \leq k)$
0	0.316	
1	0.422	
2	0.211	
3	0.047	
4	0.004	

$n = 4, p = 0.25$

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Your turn

What is the chance of **at most 1** positive test, $P(X \leq 1)$?

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k	$P(X = k)$	$P(X \leq k)$
0	0.316	0.316
1	0.422	0.738
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Your turn

What is the chance of **at most 1** positive test, $P(X \leq 1)$?

$P(X \leq 1) = 0.316 + 0.422 = 0.738$

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0	0.316	0.316
1	0.422	0.738
2	0.211	0.949
3	0.047	0.996
4	0.004	1.000

$n = 4, p = 0.25$

Your turn

What is the chance of **at most 1** positive test, $P(X \leq 1)$?

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Your turn

What is the chance of **at most 1** positive test, $P(X \leq 1)$?

$P(X \leq 1) = 0.316 + 0.422 = 0.738$

Watch the inequality:

- $P(X < 2) = P(X \leq 1) = 0.738$
- $P(X \geq 2) = 1 - P(X \leq 1) = 0.262$

In R: binomial probabilities

```
# The clinic: n = 4 patients, p = 0.25
choose(4, 1)                          # 4
dbinom(1, size = 4, prob = 0.25)      # P(X = 1)  = 0.42
pbinom(1, size = 4, prob = 0.25)      # P(X <= 1) = 0.74
1 - pbinom(1, size = 4, prob = 0.25) # P(X >= 2) = 0.26
```

d is for one value, $P(X = k)$; p adds up, $P(X \leq k)$.

Check your understanding

Your turn

A clinic tests $n = 5$ patients; each tests positive with chance $p = 0.2$.

1. What are $E(X)$ and $\mathrm{Var}(X)$?
2. What is $P(X = 0)$?
3. What is $P(X \leq 1)$?

Check your understanding

Your turn

A clinic tests $n = 5$ patients; each tests positive with chance $p = 0.2$.

1. What are $E(X)$ and $\mathrm{Var}(X)$?
2. What is $P(X = 0)$?
3. What is $P(X \leq 1)$?

1. $E(X) = 5 \times 0.2 = 1; \quad \mathrm{Var}(X) = 5 \times 0.2 \times 0.8 = 0.8$
2. $P(X = 0) = \binom{5}{0}(0.2)^0(0.8)^5 = 0.328$
3. $P(X \leq 1) = 0.328 + \binom{5}{1}(0.2)^1(0.8)^4 = 0.328 + 0.410 = 0.737$

What if we do not know the total?

Remember

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- So far we knew n and p and asked about X.

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- Can the binomial help us count a population that nobody can count directly?

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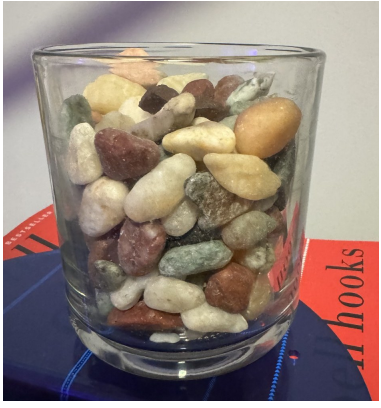
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Let’s try it with a jar of pebbles.

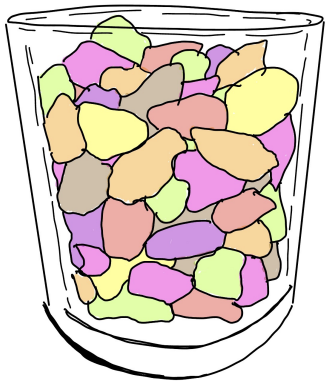
Activity: how many pebbles are in this jar?



Your turn

Without counting every pebble, how could we estimate the total?
Share one idea with your neighbor.

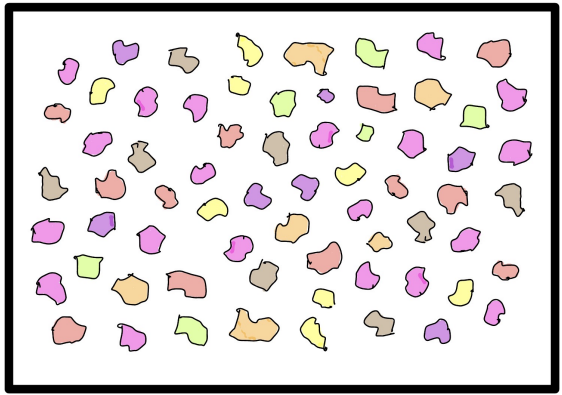
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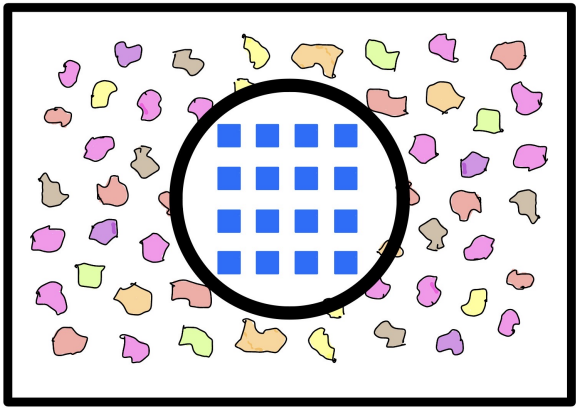
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Our plan, step by step



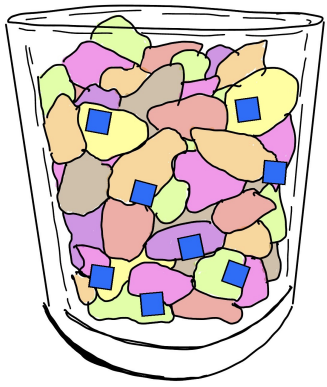
1. Toss the pebbles onto the mat.

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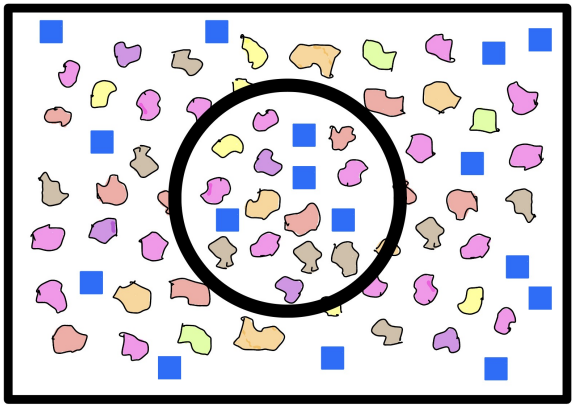
1. Toss the pebbles onto the mat.
2. Swap every pebble inside the circle for a **blue cube**. Count the swaps: M.

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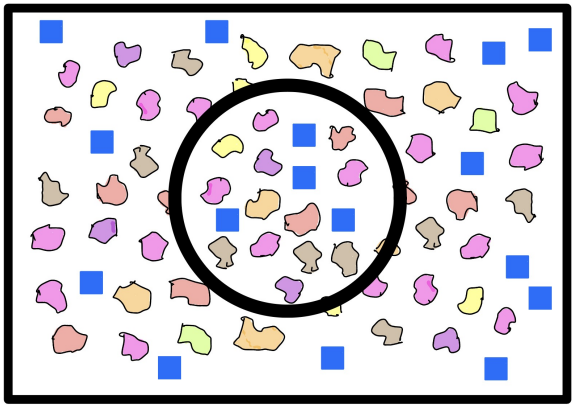
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3. Pour everything back into the jar and mix well.

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4. Toss again. Inside the circle, count all objects (n) and the cubes (k).

Our plan, step by step



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Idea: the share of cubes in the circle should be close to the share of cubes in the jar.

The jar is a binomial experiment

On the second toss, each object inside the circle is a cube (success) or a pebble (failure). Swapping keeps the jar's total at N.

In the jar	In the binomial
blue cubes swapped in	M (marked)

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objects inside the circle on the second toss	n (number of tries)
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share of the jar that is cubes	$p = M/N$

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$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \text{with } p = \frac{M}{N}$$

Your turn: how many objects are in the jar?

Your turn

One of my runs at home: I swapped in $M = 21$ cubes. On the second toss, $n = 21$ objects landed in the circle, and $k = 3$ of them were cubes.
About how many objects are in the jar?

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About how many objects are in the jar?

- In the circle, 3 of the 21 objects are cubes: **one seventh**.

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About how many objects are in the jar?

- In the circle, 3 of the 21 objects are cubes: **one seventh**.
- If the jar is well mixed, cubes should be about one seventh of the whole jar too:

$$\frac{\text{cubes in the circle}}{\text{objects in the circle}} = \frac{\text{cubes in the jar}}{\text{objects in the jar}} \quad \Longrightarrow \quad \frac{3}{21} = \frac{21}{N}$$

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- In the circle, 3 of the 21 objects are cubes: **one seventh**.
- If the jar is well mixed, cubes should be about one seventh of the whole jar too:

$$\frac{\text{cubes in the circle}}{\text{objects in the circle}} = \frac{\text{cubes in the jar}}{\text{objects in the jar}} \quad \Longrightarrow \quad \frac{3}{21} = \frac{21}{N}$$

- So 21 cubes are one seventh of the jar: $N \approx 21 \times 7 = 147$.

The binomial gives the same answer

The cubes in the circle are binomial with $n = 21$ tries and $p = M/N$. We know M but not N, so this time we do **not** know p. On average, the circle holds $E(X) = np$ cubes; we set that average equal to the k cubes we saw:

$$E(X) = n \times \frac{M}{N}$$

- M = cubes swapped in on the first toss
- n = objects inside the circle on the second toss
- k = cubes inside the circle on the second toss

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$$E(X) = n \times \frac{M}{N} \quad \Longrightarrow \quad k = n \times \frac{M}{\hat{N}} \quad \Longrightarrow \quad \hat{N} = \frac{M \times n}{k}$$

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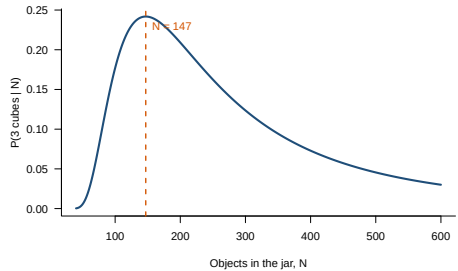
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- M = cubes swapped in on the first toss
- n = objects inside the circle on the second toss
- k = cubes inside the circle on the second toss

The **Lincoln–Petersen estimator**: the same ratio as before. Of all values of N, 147 makes the observed 3 cubes most likely.



In R: estimate the jar total

```
# One of my runs at home
M <- 21   # cubes swapped in
n <- 21   # objects in the circle
k <- 3    # cubes in the circle

M * n / k                          # Lincoln-Petersen: 147
```

From the jar to global public health

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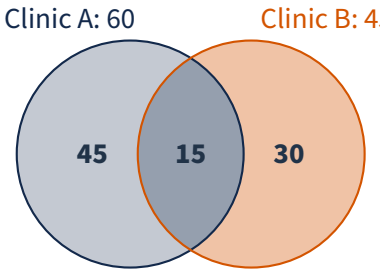
From the jar to global public health

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- The largest known HD family lives along **Lake Maracaibo**, Venezuela; their samples helped find the HD gene.
- Across Latin America, many people with HD are never diagnosed, and few countries have a registry.

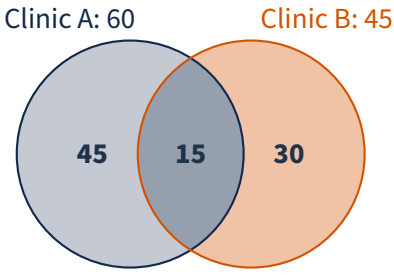


Two clinics, two lists



(Illustrative numbers.)

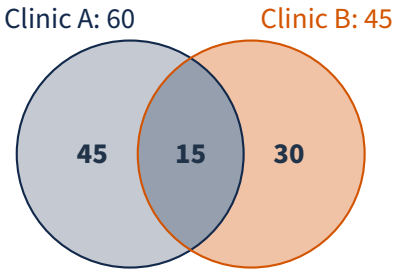
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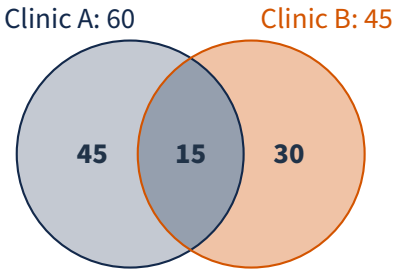
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(Illustrative numbers.)

- Clinic A has diagnosed **60** people with HD: the “blue cubes”.
- Clinic B has diagnosed **45**: the “second toss”.

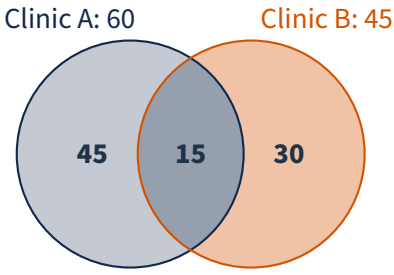
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Your turn

Estimate the number of people with HD in the region.

Answer: the same formula as the jar

In the jar	In the clinics	
M (cubes)	on A's list	60
n (in circle)	on B's list	45
k (cubes in circle)	on both	15

Answer: the same formula as the jar

In the jar	In the clinics	
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n (in circle)	on B's list	45
k (cubes in circle)	on both	15

$$\hat{N} = \frac{M \times n}{k} = \frac{60 \times 45}{15} = 180$$

Answer: the same formula as the jar

In the jar	In the clinics	
M (cubes)	on A's list	60
n (in circle)	on B's list	45
k (cubes in circle)	on both	15

$$\hat{N} = \frac{M \times n}{k} = \frac{60 \times 45}{15} = 180$$

An alternative is the **Chapman estimator**:

$$\tilde{N} = \frac{(M+1)(n+1)}{k+1} - 1 = \frac{61 \times 46}{16} - 1 \approx 174$$

Nearly the same here; the Chapman estimator matters most when k is small, and it still works when $k = 0$.

In R: two clinic lists

```
M <- 60   # on Clinic A's list
n <- 45   # on Clinic B's list
k <- 15   # on both lists

M * n / k                          # Lincoln-Petersen: 180
(M + 1) * (n + 1) / (k + 1) - 1    # Chapman: 174
```

When the estimate goes wrong

A region with exactly **180** people with HD, two clinics, 1,000 simulations. Is $\hat{N}$ too large or too small?

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Referral

Fails: independent lists

Clinic A refers its patients to Clinic B, so the overlap is too large.

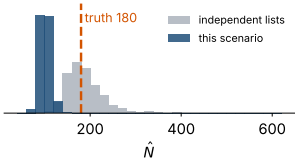
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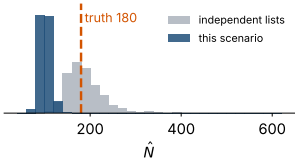
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Hard-to-reach group

Fails: same p for everyone

Rural, uninsured, or undocumented families rarely reach either clinic.



Median 101: **too small**

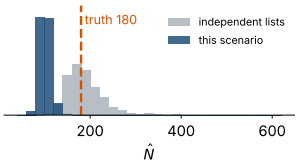
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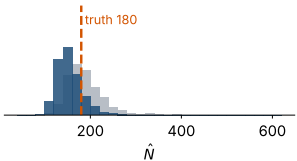


Median 101: **too small**

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Fails: same p for everyone

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Median 150: **too small**

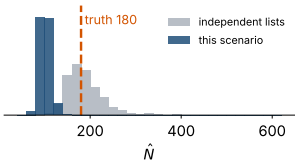
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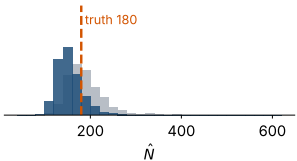


Median 101: **too small**

Hard-to-reach group

Fails: same p for everyone

Rural, uninsured, or undocumented families rarely reach either clinic.



Median 150: **too small**

Separate catchments

Fails: same p for everyone

Each clinic mostly sees people who live nearby, so the overlap is too small.

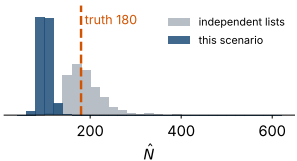
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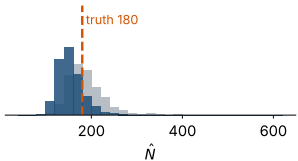


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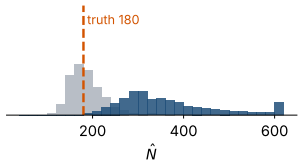


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Separate catchments

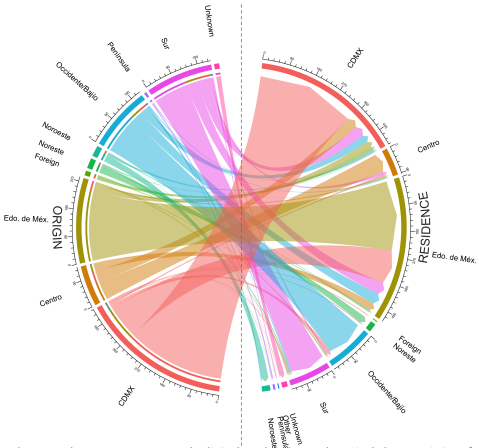
Fails: same p for everyone

Each clinic mostly sees people who live nearby, so the overlap is too small.



Median 340: **too large**

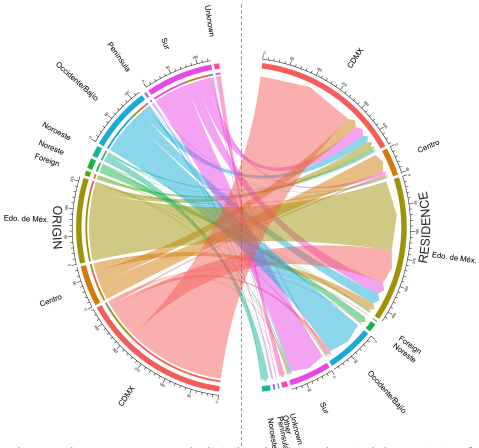
Who is hard to reach? HD in Mexico



Ochoa-Morales A, Vazquez JE, et al. Clinical, molecular, and survival characteristics of Huntington's disease in Mexico: a 30-year cohort from the National Referral Center. In preparation.

- Left: region of origin. Right: region of residence. Many people from other regions now live in Mexico City (CDMX) or Estado de México.

Who is hard to reach? HD in Mexico



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- Left: region of origin. Right: region of residence. Many people from other regions now live in Mexico City (CDMX) or Estado de México.
- Moving to Mexico City takes money. Low-income families stay in their region and may reach neither clinic: they are the **hard-to-reach group**, so $\hat{N}$ is too small.

Summary

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1. A binomial count has a fixed number of independent tries with the same chance of success p; its mean is np.

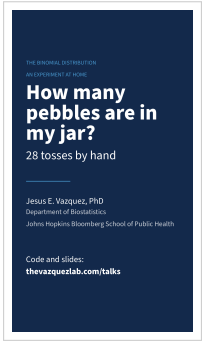
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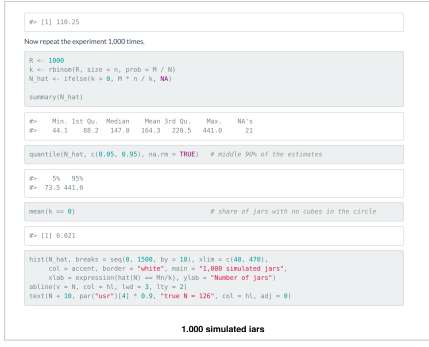
1. A binomial count has a fixed number of independent tries with the same chance of success p; its mean is np.
2. Setting np equal to the observed count turns the binomial into an estimator of a population size: $\hat{N} = Mn/k$.
3. The estimate is only as good as the assumptions. Always ask who has a lower chance of appearing on a list.

Try it yourself



A short video of my 28 tosses

youtu.be/LEBHiUieLOI



```
#> [1] 110.25
```

Now repeat the experiment 1,000 times.

```
R <- 1000
k <- rbinom(R, size = n, prob = M / N)
N_hat <- ifelse(k > 0, M * n / k, NA)

summary(N_hat)
```

```
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
#>    44.1    88.2   147.0   164.3   220.5   441.0      21
```

```
quantile(N_hat, c(0.05, 0.95), na.rm = TRUE)   # middle 90% of the estimates
```

```
#>    5%   95%
#>  73.5 441.0
```

```
mean(k == 0)                              # share of jars with no cubes in the circle
```

```
#> [1] 0.021
```

```
hist(N_hat, breaks = seq(0, 1500, by = 10), xlim = c(40, 470),
     col = accent, border = "white", main = "1,000 simulated jars",
     xlab = expression(hat(N) == M/k), ylab = "Number of jars")
abline(v = N, col = hl, lwd = 3, lty = 2)
text(N + 10, par("usr")[4] * 0.9, "true N = 126", col = hl, adj = 0)
```

1.000 simulated iars

R code for every simulation and plot

thevazquezlab.com/talks

Thank you!

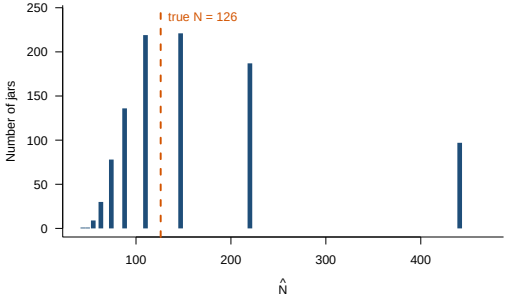
Questions?

Jesus E. Vazquez, PhD
Department of Biostatistics
Johns Hopkins Bloomberg School of Public Health

Slides available at **thevazquezlab.com/talks**

Supplementary materials

A small simulation: how much does $\hat{N}$ vary?



Let the computer fill a jar like mine with exactly **126** objects, swap in 21 cubes, take 21 objects, and compute $\hat{N}$. Repeat 1,000 times.

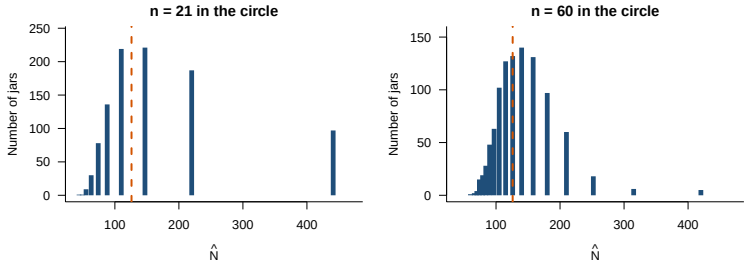
- Around the truth: the most common estimates are 147 ($k = 3$) and 110 ($k = 4$).
- Wide: the middle 90% runs from 74 to 441.
- Skewed right: a toss with few cubes in the circle gives a very large $\hat{N}$.
- Gaps: k is a whole number, so $\hat{N} = 441/k$ takes only certain values; 2% of jars had $k = 0$.

In R: simulate 1,000 jars

```
M <- 21; n <- 21; N <- 126
k_sim <- rbinom(1000, size = n, prob = M / N)
N_hat <- M * n / k_sim

summary(N_hat)
hist(N_hat)
```

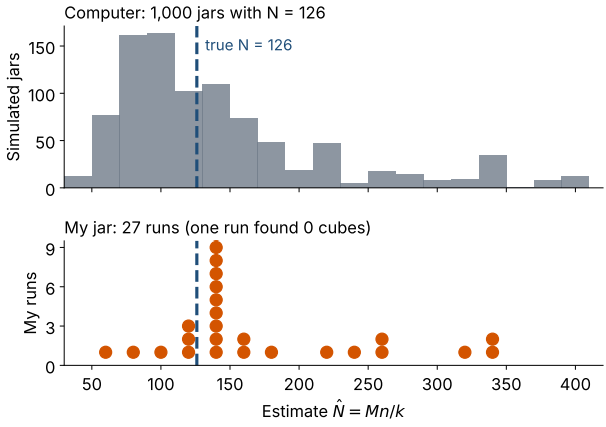
Simulation: what does a bigger second toss buy?



	$n = 21$	$n = 60$
Jars with $k = 0$ (no estimate)	21	0
Middle 90% of estimates	74 to 441	84 to 210

More objects in the circle $\Rightarrow$ more cubes found $\Rightarrow$ a less variable $\hat{N}$.

My jar at home versus the computer



My jar: $N = 126$ objects, $M = 21$ cubes. I tossed it **28** times.

- Median estimate: **147**; the computer's median is 116.
- Cubes were 14% of the objects in the circle, but 17% of the jar.
- One run found **0** cubes: $\hat{N} = Mn/0$ is undefined.

Cubes were somewhat less likely than pebbles to land in the circle, so $\hat{N}$ ran high.

Counting people is not counting pebbles

- An undercount means fewer neurologists, genetic counselors, and care services for the people already left out.
- An HD diagnosis carries information about a whole family: relatives share the risk, and stigma or insurance concerns can keep families away from clinics.
- Linking two clinic lists requires privacy protections and, often, consent.

For more depth

For MPH and MS students

- With small overlaps, the **Chapman estimator** $\tilde{N} = \frac{(M+1)(n+1)}{k+1} - 1$ has less bias (derived under the hypergeometric distribution) and works even when $k = 0$: for my run, $\tilde{N} = (22 \times 22)/4 - 1 = 120$.
- With three or more lists, log-linear models for the overlap counts (Poisson distribution) allow some dependence between lists (EPIB 652).
- Public health researchers use the same method to estimate the number of people who inject drugs or undercounted disease cases. For example, see Allen ST, et al. (2019). Estimating the number of people who inject drugs in a rural county in Appalachia. *American Journal of Public Health* 109(3):445–450. doi:10.2105/AJPH.2018.304873

Want to read more?

Capture-recapture in epidemiology

- Hook EB, Regal RR (1995). Capture-recapture methods in epidemiology: methods and limitations. *Epidemiologic Reviews* 17(2):243–264. doi:10.1093/oxfordjournals.epirev.a036192
- International Working Group for Disease Monitoring and Forecasting (1995). Capture-recapture and multiple-record systems estimation I: history and theoretical development. *American Journal of Epidemiology* 142(10):1047–1058. PubMed 7485050

Huntington disease and the Lake Maracaibo families

- Gusella JF, et al. (1983). A polymorphic DNA marker genetically linked to Huntington's disease. *Nature* 306:234–238. doi:10.1038/306234a0
- The Huntington's Disease Collaborative Research Group (1993). A novel gene containing a trinucleotide repeat that is expanded and unstable on Huntington's disease chromosomes. *Cell* 72(6):971–983. doi:10.1016/0092-8674(93)90585-E
- Wexler NS, et al. (2004). Venezuelan kindreds reveal that genetic and environmental factors modulate Huntington's disease age of onset. *PNAS* 101(10):3498–3503. doi:10.1073/pnas.0308679101
- Rawlins MD, et al. (2016). The prevalence of Huntington's disease. *Neuroepidemiology* 46(2):144–153. doi:10.1159/000443738